LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 12: Capstone: Data App

Module 4 | 90 minutes | Capstone project | Grades 6–8 · ages 11–14

Objective: Design, build, test and present a data-driven app that uses lists, procedures, validation and a menu or simulation.

Success criteria

- My app uses lists, procedures and validation.
- It produces a clear result.
- I tested with users and improved it.

Key vocabulary

- **user testing**: Watching real users try the app.
- **data structure**: How the data is organised (e.g. parallel lists).
- **minimum viable product**: The simplest version that's useful.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (90 min)

Time	Stage	What happens
10 min	**Session 1 · Plan**	User, data and procedures; teacher check.
30 min	**Session 1 · Data and engine**	Lists and main loop working.
5 min	**Session 1 · Save**	Save; note next step.
20 min	**Session 2 · Robustness and output**	Validation, menu, results.
10 min	**Session 2 · User testing**	Two users each; fix one issue.
15 min	**Session 2 · Submit & present**	Reflection, submit, 1-minute demos.

Unplugged warm-up

App pitch (5 min): explain your user and data in 30 seconds.

Teaching the idea

Assesses lists/traversal, procedures, validation, menus and data-driven design, plus planning and user testing.

Analogy: A small software product: plan, build, test with users, ship.

Live demo: Show the quiz app with a menu and validation; then extend it with a new topic list.

Questions to ask

Ask	Listen for
Who is your user?	Learner-specific.
What is in your lists?	The data structure.
What did testers struggle with?	Feedback.

Differentiation

Support
Extend the quiz engine with a menu and validation.

Stretch
Multiple topics chosen from a menu; best-score tracking.

Common misconceptions

Misconception	How to fix it
A capstone must be huge	Small, polished and robust is better.
Testing is optional at the end	Test throughout.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Data	No lists	One list, hard-coded use	Lists traversed by an engine	Well-designed data structure
Procedures	None	One, unclear	Focused procedures	Clean modular design
Robustness	Breaks easily	Some checks	Validation and catch-all	Handles all tested weird inputs

Usefulness	Unclear purpose	Works	Useful and clear	Polished for its user
Testing & presentation	None	Self-tested	Two users; change explained	Clear demo of data, code and improvements

Exit ticket
One change you made after user testing.

Homework
Have someone at home use your app and note one confusing moment.

Answer key — Kitabash activities

1. Match the app feature to the concept it uses (perform · matching)

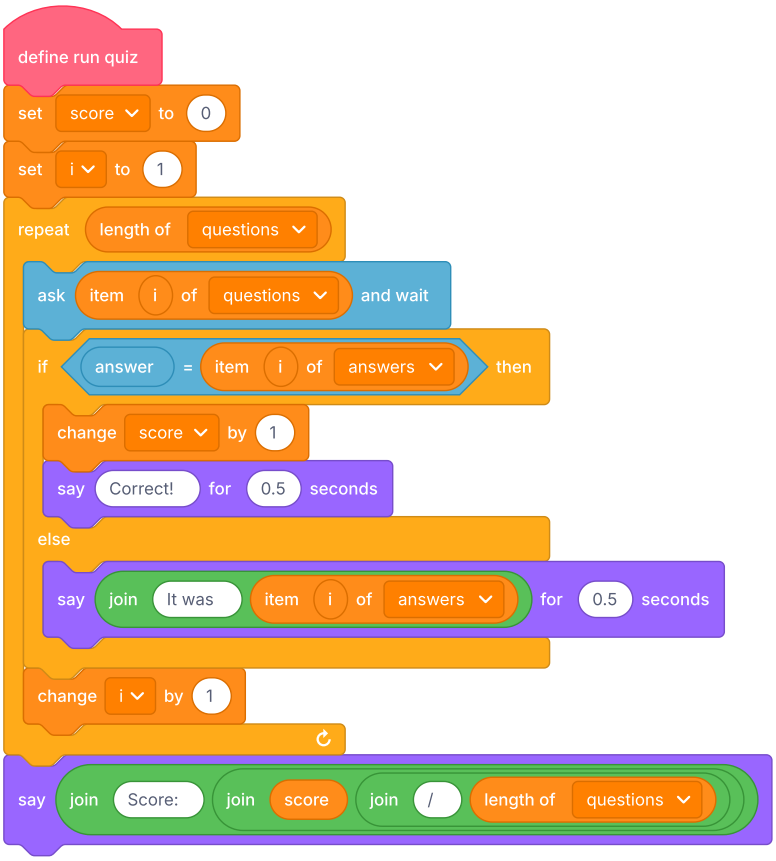
Questions and answers stored together → Parallel lists · Keep asking until a number is entered → Validation loop · A “show stats” block → Procedure · Average of all scores → Traversal with an accumulator · Choose 1, 2 or 3 → Menu with if-else

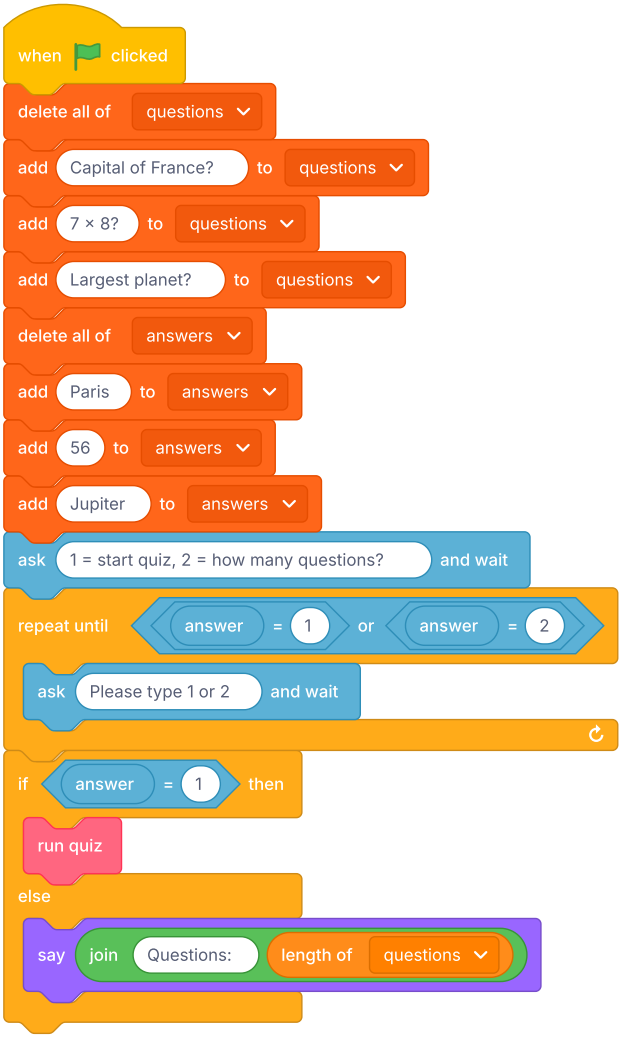
2. Capstone: Data App (probe · code lab)

Goals checked: Uses a list · Uses a procedure · Takes input · Reports a result

Example project:

Host


define run quiz
set score to 0
set i to 1
repeat length of questions
ask item i of questions and wait
if answer = item i of answers then
change score by 1
say Correct! for 0.5 seconds
else
say join It was item i of answers for 0.5 seconds
change i by 1
say join Score: join score join / length of questions


when clicked
delete all of questions
add Capital of France? to questions
add 7 × 8? to questions
add Largest planet? to questions
delete all of answers
add Paris to answers
add 56 to answers
add Jupiter to answers
ask 1 = start quiz, 2 = how many questions? and wait
repeat until answer = 1 or answer = 2
ask Please type 1 or 2 and wait
if answer = 1 then
run quiz
else
say join Questions: length of questions

Answer key — worksheet

1. **Key words:** user testing = Watching real users try the app. · data structure = How the data is organised (e.g. parallel lists). · minimum viable product = The simplest version that's useful.
2. **Match the app feature to the concept it uses:** Questions and answers stored together → Parallel lists · Keep asking until a number is entered → Validation loop · A "show stats" block → Procedure · Average of all scores → Traversal with an accumulator · Choose 1, 2 or 3 → Menu with if-else
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.